# F7 second-pass preparation

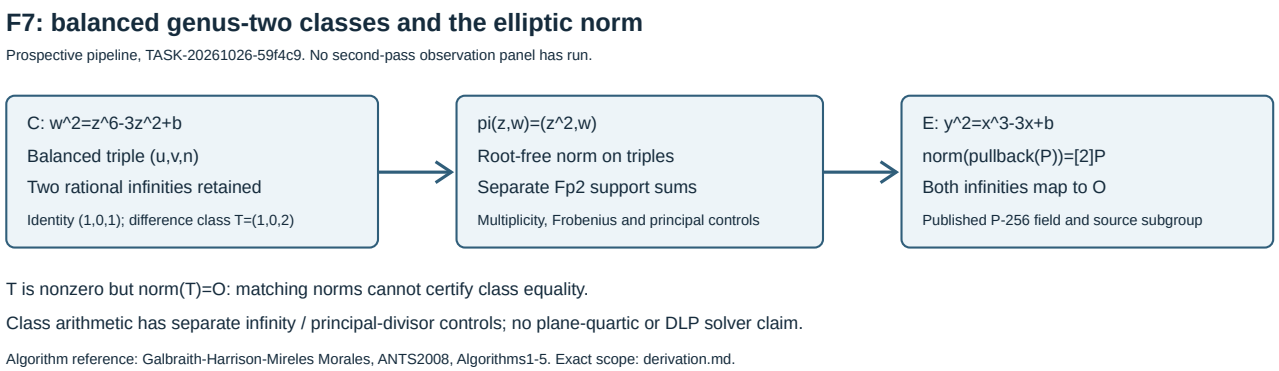


Figure 1: Editable study diagram; exact scope and evidence paths appear in the report.

26 October 2026. Task TASK-20261026-59f4c9 is prepared for balanced genus-two arithmetic and norm checks. No second-pass arithmetic panel or pilot has run. Root must release the exact source, frozen protocol and real S3 gate before observation collection.

The proposed panel constructs sixteen fixed rational C points and eight Frobenius-stable nonsplit degree-two supports, retaining every rejected draw. It executes sixteen rational add/subtract/double/cancel cases, eight nonsplit additions, eight associativity checks, and seven prior source pullbacks with balanced class addition. Principal-divisor, infinity-difference, corruption and missing-conjugate controls cover exceptions.

The candidate implements full split-model balanced triples (u,v,n) with infinity counterweights. The actual reference reconstructs supports in Fp[i] and sums their elliptic images. The reference checks norms, and cannot prove class equality in the norm kernel. The nonzero infinity-difference class with zero norm is an explicit control against that invalid inference.

See derivation.md for the balance convention and root-free norm, and literature.md for primary sources actually read. The existing Sage10.6 even-degree pair arithmetic is not used as an oracle. No plane-quartic Jacobian arithmetic, component-specific relation solver or advantage claim is supplied.

Future commands, sequential with F5 in this worker, after root's release:

```
python3 -B research/p256_factor_base_parallel_20261026/tracks/f7/run.py --out
research/p256_factor_base_parallel_20261026/tracks/f7/runs/run-01 --seed 2026102607 --release
research/p256_factor_base_parallel_20261026/runtime/releases/TASK-20261026-59f4c9.json
python3 -B research/p256_factor_base_parallel_20261026/tracks/f7/export.py
research/p256_factor_base_parallel_20261026/tracks/f7/runs/run-01 --out
research/p256_factor_base_parallel_20261026/tracks/f7/export-run-01.json
python3 tools/experiment_protocol.py check-export
research/p256_factor_base_parallel_20261026/tracks/f7/export-run-01.json --protocol
research/p256_factor_base_parallel_20261026/tracks/f7/protocol.json --verify-files
```

Each fresh run directory retains raw cases, traces, checks, rows, identities, resource diagnostics and failures. Process clocks are diagnostic, with wall_valid=false. There is no observation export or result before execution; root will render and inspect the new report PDFs from actual recorded data.